



US 20250282248A1

(19) **United States**

(12) **Patent Application Publication**
Alarcon et al.

(10) **Pub. No.: US 2025/0282248 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **ELECTRIC VEHICLE CHARGING
INFRASTRUCTURE AND METHOD FOR
CHARGING AN ELECTRIC VEHICLE**

(52) **U.S. Cl.**
CPC **B60L 53/665** (2019.02); **B60L 53/305**
(2019.02)

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(57) **ABSTRACT**

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An electric vehicle charging infrastructure is provided which includes an electric vehicle supply equipment (EVSE) for charging an electric vehicle (EV) and a policy configurator. The EVSE includes a human-machine interface (HMI) for interacting with a user, a payment subsystem for processing payments, a charging subsystem for charging the EV, and a master event processor. The policy configurator is configured a policy set and for sending the policy set to the master event processor. The master event processor receives event data from the HMI and the payment subsystem and in response to the received event data, controls operation of the HMI and of the payment subsystem according to a workflow based on the policy set. The workflow specifies a respective action for enabling a charging operation in response to the at least one non-critical anomaly event.

(21) Appl. No.: **19/072,251**

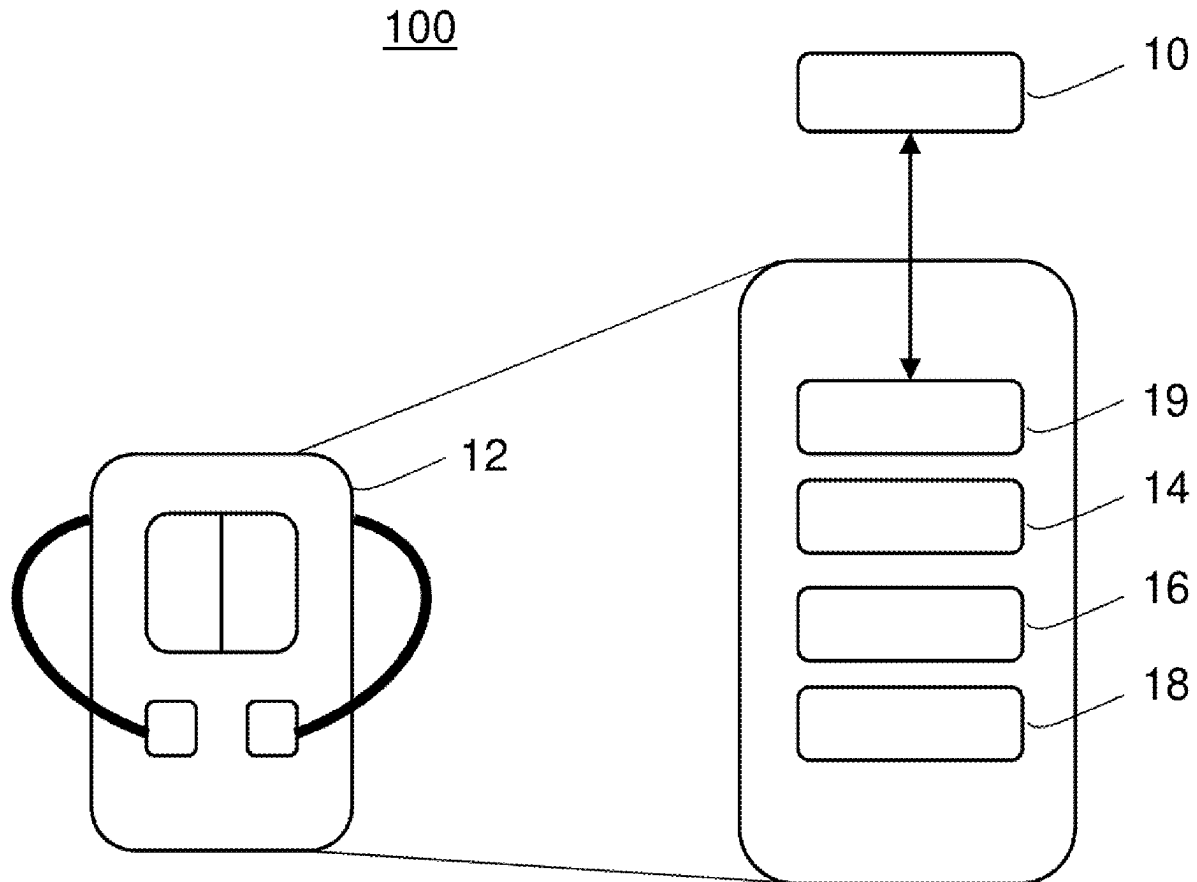
(22) Filed: **Mar. 6, 2025**

(30) **Foreign Application Priority Data**

Mar. 6, 2024 (EP) 24161713.3

Publication Classification

(51) **Int. Cl.**
B60L 53/66 (2019.01)
B60L 53/30 (2019.01)



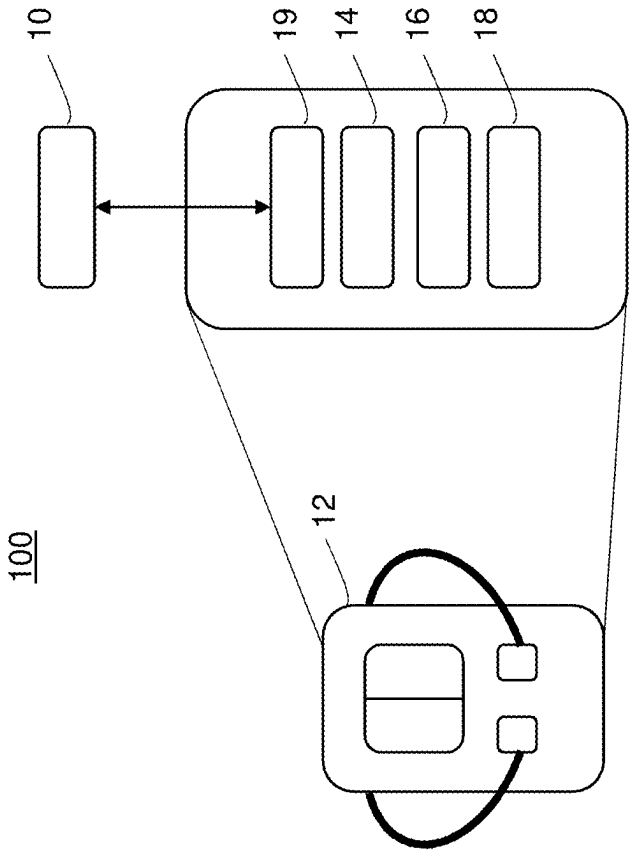
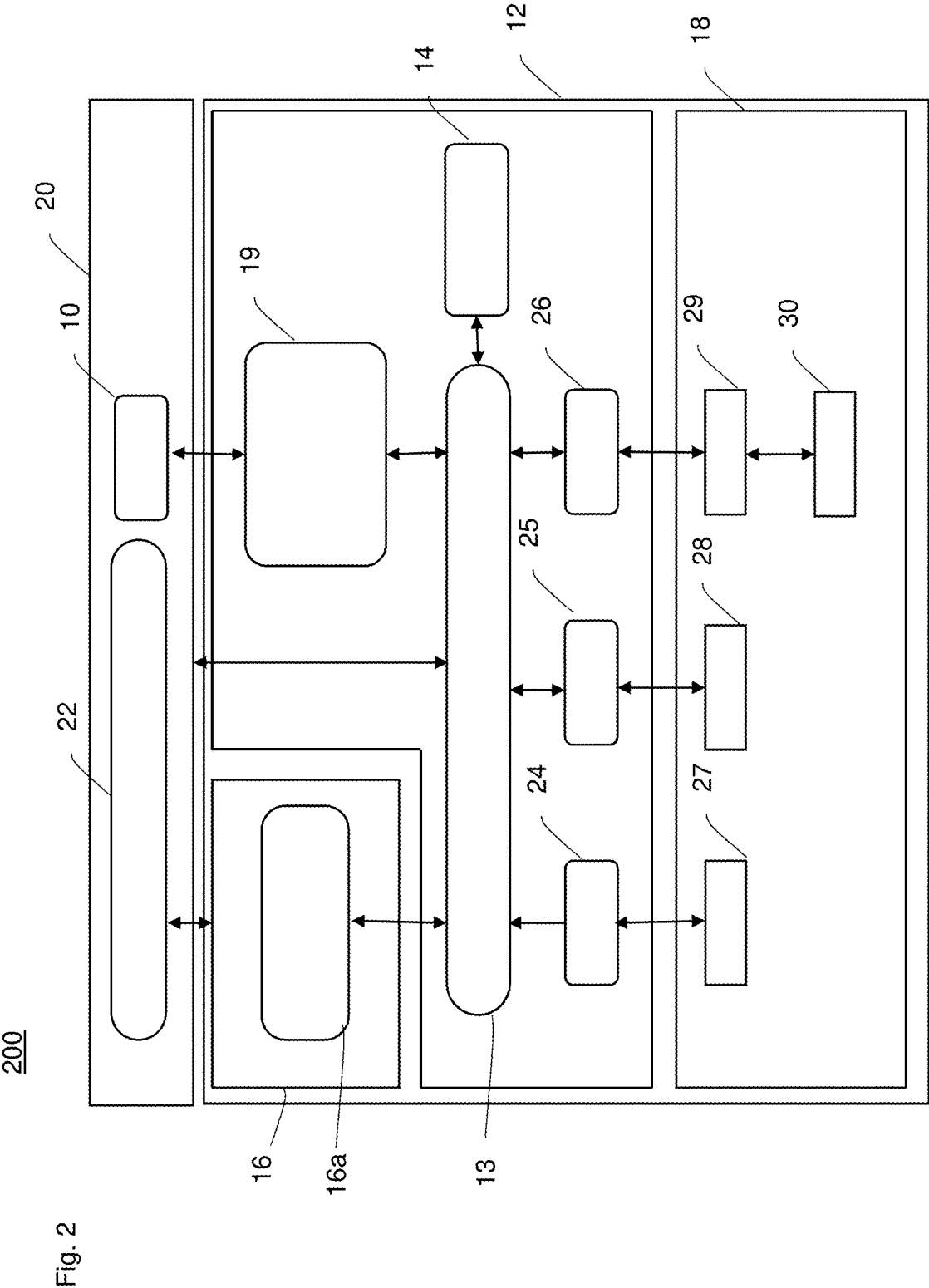


Fig. 1



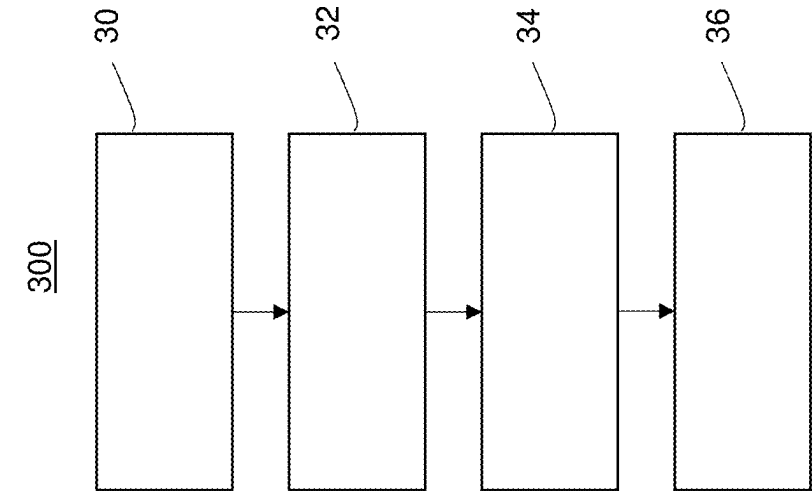


Fig. 3

ELECTRIC VEHICLE CHARGING INFRASTRUCTURE AND METHOD FOR CHARGING AN ELECTRIC VEHICLE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to European Patent Application No. 24161713.3 filed on Mar. 6, 2024, and titled “ELECTRONIC VEHICLE CHARGING INFRASTRUCTURE AND METHOD FOR CHARGING AN ELECTRIC VEHICLE”, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] Aspects of the present disclosure generally relate to an electric vehicle charging infrastructure, in particular to an electric vehicle charging infrastructure allowing the flexible adaptation of workflows. Further aspects relate to a corresponding method for charging an electric vehicle.

BACKGROUND

[0003] Efforts have been made to standardize and ensure interoperability between Electric Vehicle Charging Equipment (EVSE) and Electric Vehicles (EVs) for EV charging, in particular open standards for electric vehicle charging infrastructure, such as, for instance the Open Charge Point Protocol (OCPP). Still, the interoperability remains imperfect. These efforts are further complicated by the large number of possible combinations and scenarios, and by the fragmented landscape of electric vehicle charging infrastructure where components and/or services are provided by various companies. Each company may define their own backend-parameters even within a given standard. This can lead to unforeseen compatibility issues. For such reasons, attempts to charge an electric vehicle may be cancelled without charging success.

[0004] Another factor contributing to the above issues is the interaction of the electric vehicle charging infrastructure with a user. The electric vehicle charging infrastructure might provide for a static workflow, expecting a fixed sequence of user actions, while a user might behave more intuitively or like he is accustomed to from another charging infrastructure from a different provider. This might lead to a mismatch between expected and actual user action.

[0005] One way to reduce the above issues is testing. Indeed, testing of electric vehicle charging infrastructure can reduce the number of the above issues. But not all scenarios can be tested reliably in advance, so that there may remain scenarios resulting in unwanted cancellation of charging attempts despite operation equipment of an electric vehicle charging infrastructure. Thus, there is still room to further improve the charging success rate of the charging infrastructure.

BRIEF DESCRIPTION

[0006] Technical terms are used in their common sense. If a specific meaning is conveyed to certain terms, definitions of terms will be given in the following in the context of which the terms are used.

[0007] According to an aspect, an electric vehicle charging infrastructure is provided. The electric vehicle charging infrastructure includes an electric vehicle supply equipment (EVSE) for charging an electric vehicle (EV) and a policy

configurator. The EVSE includes a human-machine interface (HMI) for interacting with a user, a payment subsystem for processing payments, a charging subsystem for charging the EV, and a master event processor. The HMI, the payment subsystem and the charging subsystem are adapted for issuing event data to the master event processor.

[0008] The policy configurator is adapted for configuring a policy set and for sending the policy set to the master event processor. The master event processor is adapted for receiving the event data from the HMI, the payment subsystem, and in some embodiments the charging subsystem, and in response to the received event data, controlling operation of the HMI, of the payment subsystem, and in some embodiments of the charging subsystem according to a workflow based on the policy set.

[0009] According to an aspect, a method for charging an EV by an EVSE is provided. The method includes configuring a policy set; sending the policy set to a master event processor of the EVSE; receiving, by the master event processor, event data from an HMI, a payment subsystem, and in some embodiments a charging subsystem, of the EVSE; and controlling, in response to the received event data, operation of the HMI, payment subsystem, and in some embodiments the charging subsystem according to a workflow based on the policy set.

[0010] The above-defined method may be carried out by any configuration of the EVSE described above.

[0011] According to an aspect, the method may include receiving, by the master event processor, event data indicative of an EV being connected to the EVSE for charging the EV by the EVSE. The event data may be indicative of a user's intention to charge the EV by the EVSE.

[0012] Aspects of the present disclosure allow for an improved electric vehicle charging infrastructure allowing adaptable workflows. In particular, by executing overall control according to a workflow based on the policy set and taking account events from various relevant sources, a highly coordinated charging action is enabled which allows successful charging operation even in the face of minor errors or inconsistencies. Such errors and inconsistencies may include incomplete authorization, inconsistencies of backend messaging, uncritical charger errors, uncritical protocol errors, human user errors, and/or payment errors. By providing an overall workflow controlled by the master event processor as well as a flexibly configurable policy set guiding this workflow, it is made possible to handle such errors and inconsistencies gracefully for ensuring a successful overall charging operation. Further, a highly flexible adaptation of the policy set is made possible, which allows adapting the policy set in response to new circumstances or new experiences.

[0013] Thus, aspects of the present disclosure enable flexible, and possibly automated, adjustment of policies based on field operational data (e.g., event logs). Compared to regular software or firmware updates that require strict testing release cycle management, aspects of the present disclosure allow a more flexible and adaptive updating. Thereby, it is possible to perform instant deployment, fast iterations, tests, selective deployment, and experimentation. In contrast, full software or firmware updates, which cannot be carried out with the same level of flexibility and confidence as the updating or configuration of policy sets only (which does not require an overall software update).

[0014] Further advantages, features, aspects and details that can be combined with embodiments described herein are evident from the dependent claims, the description and the drawings.

BRIEF DESCRIPTION OF DRAWINGS

[0015] The details will be described in the following with reference to the figures.

[0016] FIG. 1 is a schematic circuit diagram of an electric vehicle charging infrastructure according to an embodiment.

[0017] FIG. 2 is a schematic diagram of an electric vehicle charging infrastructure according to an embodiment.

[0018] FIG. 3 is a schematic diagram of a method for charging an EV.

DETAILED DESCRIPTION

[0019] In the following, embodiments are set forth to describe specific examples presented herein. The person skilled in the art will recognize that one or more other examples and/or variations of these examples may be practiced without all the specific details outlined below. Also, well known features may not be described in detail so as not to obscure the description of the examples herein. For the ease of illustration, like reference numerals are used in different figures to refer to the same elements or additional instances of the same element. Features illustrated or described as part of one embodiment can be used on or in conjunction with any other embodiment to yield yet a further embodiment. The figures represent schematic diagrams and do not necessarily represent dimensions or measurements of real embodiments and/or are not necessarily drawn to scale.

[0020] Referring now to the drawings, FIG. 1 presents a schematic circuit diagram of an electric vehicle charging infrastructure 100 according to an embodiment. The electric vehicle charging infrastructure 100 includes a policy configurator 10 and an electric vehicle supply equipment (EVSE) 12 for charging an electric vehicle (EV). The EVSE 12 includes a human-machine interface (HMI) 14 for interacting with a user, a payment subsystem 16 for processing payments, a charging subsystem 18 for charging the EV, and a master event processor 19.

[0021] The policy configurator 10 may be integrated in the EVSE 12 or provided as a separate unit remote from the EVSE 12. The policy configurator 10 may communicate with the master event processor 19, possibly over a network.

[0022] FIG. 2a is a further schematic diagram of the electric vehicle charging infrastructure 200, which may be the same as infrastructure 100 described in connection with FIG. 1. The electric vehicle charging infrastructure 200 includes the components described with reference to FIG. 1, in particular the EVSE 12 and the policy configurator 10. The EVSE includes the HMI 14, the payment subsystem 16, the charging subsystem 18, and the master event processor 19 as already shown in FIG. 1.

[0023] The master event processor 19, the payment subsystem 16 and the charging subsystem 18 may be coupled to each other by an event bus 13. The HMI 14 may also be coupled to the event bus, either as a separate unit as shown in FIGS. 1 and 2, or jointly with the HMI 14 as an integrated HMI event master unit encompassing the HMI 14 and the master event processor 19. In this case, the integrated HMI event master unit may be coupled to the event bus by a

common interface, whereas the HMI and the master event processor 19 may communicate directly with each other.

[0024] The payment subsystem 16 and the charging subsystem 18 may be provided as separate modules from the master event processor 19 (or from the HMI event master unit) and may be communicatively coupled to the master event processor 19 via an event bus 13. The event bus 13 acts like a broker forwarding events from one entity to another. It may be directly connected to the charging subsystem 18, or, as depicted in FIG. 2a connected via respective agents 24, 25, and 26. The agents 24, 25, and 26 may interface with physical entities of the charging subsystem 18.

[0025] Such physical entities may include, for instance, a RFID reader 27, LEDs, LED strips, buttons, gun holder sensors, all referenced to with numeral 28, and power electronics, batteries and/or a charging station controller, all referenced to with numerals 29 and 30.

[0026] In the embodiment of FIG. 2, the electric vehicle charging infrastructure 200 further includes a cloud backend 20, and the policy configurator 10 belongs to the cloud backend 20. The cloud backend may be based on the OCPP protocol. Although in FIG. 2a the policy configurator 10 is hosted by the cloud backend 20, it may also be provided in any other manner.

[0027] The cloud backend may integrate other functionalities as well. For example, as shown in FIG. 2a, the cloud backend 20 may be a charge point operator cloud backend. The charge point operator cloud backend 20 may also provide for a cloud-based payment backend 22. The cloud-based payment backend 22 communicates with the EVSE's payment subsystem 16, e.g., directly (i.e., bypassing the event bus 13) and/or via the event bus 13. The payment subsystem 16 may include a payment HMI 16a. The payment HMI 16a may provide for user interaction for self-service payment.

[0028] The cloud backend 20 (specifically the cloud-based payment backend 22 and/or the policy configurator 10) may be communicatively coupled to the event bus 13.

[0029] The policy configurator 10 is communicatively coupled to the master event processor 19, e.g., for receiving event data, specifically log data. The charge point operator cloud backend 20 and other components of the EVSE (such as the HMI 14, the payment subsystem 16, the charging subsystem 18 the master event processor 19, and/or and integrated HMI event master unit) may communicate through the event bus 13, e.g., using a Transport Layer Security (TLS) protocol.

[0030] FIG. 3 schematically illustrates a method 300 for charging an EV by an EVSE. The method 300 includes configuring 30 a policy set; sending 34 the policy set to a master event processor of the EVSE; receiving 34, by the master event processor, event data from an HMI, a payment subsystem, and a charging subsystem of the EVSE; controlling 36, in response to the received event data, operation of the HMI, of the payment subsystem, and of the charging subsystem according to a workflow based on the policy set. The method may be carried out by any configuration of the charging infrastructure described herein. Further details regarding the EV charging infrastructure and its operation are described in the following.

[0031] Next, general optional aspects of the present disclosure are further described. Here, unless excluded, any aspect may be combined with any other aspect described

herein and may apply to any of the previously described embodiments of the present disclosure described herein, without being limited to such embodiment.

[0032] According to an aspect, the policy configurator of the electric vehicle charging infrastructure may be adapted for receiving user input log data from the HMI, receiving payment log data from the payment subsystem, receiving charging log data from the charging subsystem, and for configuring the policy set based on the received user input log data, payment log data, and charging log data. Thereby, particularly responsive adaptation of the policy set in response to the received log data, and thereby a fast feedback loop for updating the policy set in response to new circumstances or experiences is enabled.

[0033] The input log data, the payment log data, and the charging log data may be based on respective event data from the HMI, the payment subsystem, and the charging subsystem. Thereby, the policy configurator is adapted for configuring the policy set based on the respective event data from the HMI, the payment subsystem, and the charging subsystem.

[0034] According to an aspect, the input log data and/or the payment log data, and/or the charging log data may be provided to the EVSE and thus to the policy configurator by tokens. Thereby, an effective communication of the log data is enabled. The HMI, the payment subsystem, the charging subsystem, and the event processor may, for example, communicate using a Message Queuing Telemetry Transport (MQTT) protocol.

[0035] According to an aspect, the policy configurator of the electric vehicle charging infrastructure may be adapted for configuring the policy set based on event data, in particular wherein the policy configurator is configured for configuring the policy set using machine learning methods. The machine learning methods may make use of supervised algorithms or unsupervised algorithms or a combination thereof. Thereby, a particularly responsive adaptation to new circumstances and a fast and efficient feedback loop for improving operation of the EVSE may be enabled.

[0036] According to an aspect, the policy set may define the workflow by specifying, for predefined events, respective actions to be triggered in response to the predefined events. The actions may include one or more of updating a status variable, sending a command to at least one of the HMI, the payment subsystem, and the charging subsystem; updating a charging parameter; simulating a user input to the HMI; simulating an EV input to the charging subsystem; overriding a warning message; updating Human Machine Interface elements such as elements displayed on the HMI display and possible input options.

[0037] The policy set may include defining at least one non-critical anomaly event, the workflow may specify a respective action for enabling (e.g., continuing) a charging operation in response to the at least one non-critical anomaly event. The at least one non-critical anomaly event may indicate at least one of: an incomplete authorization; a messaging inconsistency; a charger error; an electric vehicle protocol error; and/or a payment error.

[0038] The charging operation may be enabled in a reduced manner, e.g., with a reduced power such as minimal charging. This may allow keeping a charging operation at least nominally running with limited (minimal) power consumption. Thereby, it becomes possible to continue the

charging process even in case of an anomaly event, e.g., if a warning message is issued, without compromising safety and reliability.

[0039] Thereby, possible non-optimal workflow paths (unsuccessful or non-optimal charging operation in spite of an intent to charge and the objective possibility to charge) can be avoided that may otherwise arise during the operation of a charger. Potential causes for such non-optimal workflow paths can be dynamic in nature, and are very difficult to predict and observe in a lab environment.

[0040] Examples of noncritical anomaly events include an incomplete (pending) authorization and/or minor authorization errors or warnings; inconsistencies of the backend (e.g., OCPP Backend) messaging; charger errors; vehicle protocol errors; user (human) errors; and/or payment errors. By recognizing such anomaly events as uncritical, the master event processor may address these issues (e.g., by overriding or working around them, or by allowing a reduced operation, e.g., with minimal charging power and/or other limitations), and thereby ensure a graceful handling of the anomaly and a successful outcome of the charging operation.

[0041] For instance, once a charging subsystem of an electric vehicle charging infrastructure is connected to an electric vehicle, a user authorization within a predetermined time period may be expected. If the user fails to authorize within this time period, the EV and/or the EVSE may cancel the charging attempt due to an incomplete authorization despite the user's attempt to charge the vehicle. The policy configurator may configure the policy set such that the user's intent (charging) is recognized and the EVSE is enabled to charge the vehicle.

[0042] The policy configurator may be configured for configuring the policy set to include decision alternatives, and wherein the event processor may be configured for implementing the decision alternatives within the workflow. The event processor may be configured to approximate an event to match one of the decision alternatives of the policy set thereby implementing the workflow. Approximating an event may comprise recognizing an intent from any component of the EVSE and/or a user based on the received events.

[0043] The policy set may further include defining at least one critical anomaly event, the workflow may specify a respective action for disabling (e.g., abandoning) a charging operation in response to the critical anomaly event. By a differentiation between uncritical and critical anomaly events, and by defining different reactions to both, safe operation is made possible.

[0044] Next, further detailed aspects relating to the system architecture are described.

[0045] According to an aspect, the master event processor and the HMI may be provided as an integrated module, e.g., jointly coupled to an event bus via a joint interface. The present disclosure is not limited thereto, and the master event processor and the HMI may alternatively be provided as separate modules within the EVSE. Still, providing the master event processor and the HMI as an integrated module can be advantageous, since the interaction between master event processor and HMI is particularly close. Specifically, the HMI may receive a user input indicative of a user's intent, and the master event processor may be enabled to establish a workflow that considers the user's intent. The workflow may also include simulating a user input and/or correcting a user input in order to remove inconsistencies or

roadblocks to the workflow. Due to these close interactions, providing the master event processor and the HMI may be as an integrated module can be advantageous.

[0046] According to an aspect, within the electric vehicle charging infrastructure, the payment subsystem and the charging subsystem may be provided as separate modules from the master event processor and may be coupled to the master event processor via an event bus.

[0047] In addition to the (primary) HMI described so far and further below, the payment subsystem may include a payment HMI (second HMI). The payment HMI may provide for self-service payment. The payment HMI may provide for payments, resource bookings as for instance reserving the EVSE for a later point in time, and/or redemption of digital charging vouchers. The payment HMI may be provided separately from the primary HMI.

[0048] According to an aspect, the policy configurator may be located remotely from the EVSE and may be connected to the master event processor over a data network, in particular wherein the policy configurator may be hosted in a cloud. According to an aspect, the policy configurator may be integrated with a Charge Point Operator Backend. According to an aspect, the Backend may be a cloud backend, in particular a cloud backend based the Open Charge Point Protocol (OCPP).

[0049] Generally, the cloud may include the electric vehicle charging infrastructure's backend. The electric vehicle charging infrastructure's backend may be a charge point operator cloud backend and/or based on the Open Charge Point Protocol (OCPP).

[0050] The disclosed systems and methods are not limited to the specific embodiments described herein. Rather, components of the systems or steps of the methods may be utilized independently and separately from other described components or steps.

[0051] This written description uses examples to disclose various embodiments, which include the best mode, to enable any person skilled in the art to practice those embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope is defined by the claims and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

1-12. (canceled)

13. An electric vehicle charging infrastructure, comprising:

- a policy configurator; and
 - an electric vehicle supply equipment (EVSE) for charging an electric vehicle (EV),
- the EVSE comprising:
- a human-machine interface (HMI) configured to enable interaction with a user;
 - a payment subsystem configured to process payments;
 - a charging subsystem configured to charge the EV; and
 - a master event processor,
- wherein each of the HMI, the payment subsystem, and the charging subsystem are configured to issue respective event data to the master event processor,

wherein the policy configurator is configured to generate a policy set and send the policy set to the master event processor; and

wherein the master event processor is configured to:

- receive the event data from the HMI and the payment subsystem; and
- in response to the received event data, control operation of the HMI, and of the payment subsystem according to a workflow based on the policy set,

wherein the workflow specifies a respective action for enabling a charging operation in response to the at least one non-critical anomaly event.

14. The electric vehicle charging infrastructure of claim 13, wherein the master event processor is configured to:

- receive the event data from the HMI, the payment subsystem, and the charging subsystem; and
- in response to the received event data, control operation of the HMI, the payment subsystem, and the charging subsystem according to the workflow based on the policy set.

15. The electric vehicle charging infrastructure of claim 13, wherein the policy configurator is configured to receive at least one user-interaction log data selected from:

- user input log data from the HMI; and
- payment log data from the payment subsystem.

16. The electric vehicle charging infrastructure of claim 15, wherein the input log data and the payment log data are based on respective event data from the HMI and the payment subsystem.

17. The electric vehicle charging infrastructure of claim 13, wherein the policy configurator is configured to generate the policy set based on event data.

18. The electric vehicle charging infrastructure of claim 13, wherein the policy set defines the workflow by specifying, for predefined events, respective actions to be triggered in response to the predefined events.

19. The electric vehicle charging infrastructure of claim 13, wherein the policy set comprises:

- defining the at least one non-critical anomaly event, the workflow specifying a respective action configured to enable a charging operation in response to the at least one non-critical anomaly event; and
- defining at least one critical anomaly event, the workflow specifying a respective action for disabling a charging operation in response to the critical anomaly event.

20. The electric vehicle charging infrastructure of claim 13, wherein the at least one non-critical anomaly event indicates at least one of:

- an incomplete authorization;
- a messaging inconsistency;
- a charger error;
- a Vehicle protocol error; and
- a payment error.

21. The electric vehicle charging infrastructure of claim 13, wherein the one non-critical anomaly event is a user-interaction anomaly.

22. The electric vehicle charging infrastructure of claim 21, wherein the non-critical anomaly event is comprises at least one of a user input anomaly and a payment anomaly.

23. The electric vehicle charging infrastructure of claim 13, wherein the workflow specifies a reduced-power charging operation in response to the at least one non-critical anomaly event.

24. The electric vehicle charging infrastructure of claim **13**, wherein the master event processor and the HMI are provided as an integrated module.

25. The electric vehicle charging infrastructure of claim **24**, wherein the integrated module is separate from a charging subsystem and configured to charge the EV.

26. The electric vehicle charging infrastructure of claim **13**, wherein the payment subsystem and the charging subsystem are provided as separate modules from the master event processor and are coupled to the master event processor via an event bus.

27. The electric vehicle charging infrastructure of claim **13**, wherein the policy configurator is located remotely from the EVSE and connected to the master event processor over a data network.

28. A method for charging an electric vehicle (EV) by an electric vehicle supply equipment (EVSE), the method comprising:

configuring a policy set;

sending the policy set to a master event processor of the EVSE;

receiving, by the master event processor, event data from a human-machine interface (HMI), and a payment subsystem of the EVSE;

controlling, in response to the received event data, operation of the HMI, and of the payment subsystem according to a workflow based on the policy set,

wherein the workflow specifies a respective action configured to enable a charging operation in response to the at least one non-critical anomaly event.

29. The method of claim **28**, the method further comprising:

receiving at least one user-interaction log data selected from user input log data from the HMI and payment log data from the payment subsystem.

30. The method of claim **28**, further comprising:

defining at least one non-critical anomaly event, the workflow specifying a respective action configured to enable a charging operation in response to the at least one non-critical anomaly event; and

defining at least one critical anomaly event, the workflow specifying a respective action for disabling a charging operation in response to the critical anomaly event.

31. The method of claim **28**, wherein the method is performed by the EVSE, the EVSE comprising:

the HMI, wherein the HMI is configured to enable interaction with a user;

the payment subsystem, wherein the payment configured to process payments;

a charging subsystem to charge the EV; and

the master event processor,

wherein each of the HMI, the payment subsystem and the charging subsystem are configured to issue respective event data to the master event processor,

wherein the policy configurator is configured to generate the policy set and send the policy set to the master event processor, and

wherein the master event processor configured to:

receive the event data from the HMI and the payment subsystem; and

in response to the received event data, control operation of the HMI, and of the payment subsystem according to a workflow based on the policy set,

wherein the workflow specifies the respective action for enabling the charging operation in response to the at least one non-critical anomaly event.

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